

Practice problems #5

1. Let $T : \mathbf{P}_2 \rightarrow \mathbf{P}_3$ be the transformation that maps a polynomial $p(t)$ into the polynomial $(t + 5)p(t)$.
 - (a) Find the image of $p(t) = 2 - t + t^2$.
 - (b) Show that T is a linear transformation.
 - (c) Find the matrix for T relative to the standard bases for $\mathbf{P}_2$ and $\mathbf{P}_3$.

Hints or Answers on Page 2.

#1(a): $10 - 3t + 4t^2 + t^3$

#1(b): Show!

#1(c):

$$\begin{bmatrix} 5 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}$$